Ondřej Kubů

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PROFILE

Severo Ochoa postdoc at Institute of Mathematical Sciences (ICMAT-CSIC), Madrid moving into AI safety and general-purpose AI (GPAI) risk, with 8 peer-reviewed publications in integrable systems and differential geometry. Volunteer with PauseAI CZ, z. s. Available from late 2026; willing to relocate.

AI SAFETY & POLICY ENGAGEMENT

Volunteer, PauseAI CZ, z. s. (2026–present). Author of the Czech-language weekly newsletter *Pozastavení nad AI* ("Pondering AI"): 7 regular issues and a special edition to date, tracking frontier-AI capabilities and EU AI Act implementation for a Czech-speaking audience.

EMPLOYMENT

Dec 2024 – Nov 2026	**Postdoc (Severo Ochoa)**, Instituto de Ciencias Matemáticas (ICMAT), CSIC, Madrid, Spain. Research on Haantjes algebras for separation of variables of (super)integrable systems. **Supervisor:** P. Tempesta
Sep 2024 – Nov 2024	**Research Associate**, Faculty of Nuclear Sciences and Physical Engineering (FNSPE), Czech Technical University in Prague (CTU)

EDUCATION

2020 – 2024	**Ph. D.**, FNSPE CTU, Mathematical Engineering / Mathematical Physics. **Thesis:** *Integrability and superintegrability in the presence of magnetic fields: separable and nonseparable systems.* **Supervisor:** L. Šnobl; **Co-supervisor:** A. Marchesiello. *Rector's Award for Excellent Doctoral Thesis (2025).* **Grade:** 1.00 GPA (Excellent); Oral defence Grade 1.
2018 – 2020	**Master's Degree with Honors (Ing.)**, FNSPE CTU, Mathematical Physics. **Thesis:** *Integrable and superintegrable systems of cylindrical type in magnetic fields.* **Supervisor:** L. Šnobl; **Consultant:** P. Winternitz. **Grade:** A (GPA 1.31); thesis and final exam both Grade A.
2015 – 2018	**Bachelor's Degree with Honors**, FNSPE CTU, Mathematical Engineering / Mathematical Physics. **Grade:** A (GPA 1.41); thesis and final exam both Grade A.

AWARDS

Rector's Award for Excellent Doctoral Thesis, CTU (2025)

Josef Hlávka Award for the best students and graduates of public universities in Prague (2023)

Milan Odehnal Award (2024) – honorable mention (Czech Physical Society)

Dean's Award for the 2nd best Master Thesis (2021)

INTERNATIONAL EXPERIENCE

Spring 2023	**ERASMUS+ research exchange**, Universidad Complutense de Madrid. **Supervisor:** P. Tempesta
Fall 2019	**ERASMUS+ research exchange**, Université de Montréal. **Supervisor:** P. Winternitz

TEACHING

2021 – 2024 **Assistant lecturer**, Dept. of Physics, FNSPE CTU. Seminars in Theoretical Physics 2, Lie Algebras and Groups, and Geometrical Methods in Physics; exercises and consultations for ERASMUS+ students.

SKILLS

Languages: Czech (native), English (C1–C2), German (B2/C1), French (B2), Spanish (A2)

Software: Maple, LaTeX; author of an open Maple package for Nijenhuis/Haantjes torsion computation (2026)

PUBLICATIONS

Preprints

O. Kubů, D. Latini. **Haantjes algebras, Zernike system and separation of variables**, under review. arXiv:2605.23748 (2026)

Journal articles

O. Kubů, P. Tempesta. **Stäckel and Eisenhart geometries, symplectic-Haantjes manifolds and gravitation**, accepted, *Proc. R. Soc. A*; in press. DOI: 10.1098/rspa.2025.1103. arXiv:2509.19950

O. Kubů, D. Reyes, P. Tempesta, G. Tondo. Hamiltonian integrable systems in a magnetic field and symplectic-Haantjes geometry. *Proc. R. Soc. A* **480** (2024) 20240076

O. Kubů, A. Marchesiello, L. Šnobl. Integrable systems with velocity-dependent potentials: generalized parabolic cylindrical case. *J. Phys. A* **57** (2024) 235203

O. Kubů, A. Marchesiello, L. Šnobl. New classes of quadratically integrable systems in magnetic fields: the generalized cylindrical and spherical cases. *Ann. Phys.* **451** (2023) 169264

O. Kubů, L. Šnobl. Cylindrical first-order superintegrability with complex magnetic fields. *J. Math. Phys.* **64** (2023) 062101

M. F. Hoque, O. Kubů, A. Marchesiello, L. Šnobl. New classes of quadratically integrable systems with velocity dependent potentials: non-subgroup type cases. *Eur. Phys. J. Plus* **138** (2023) 845

O. Kubů, A. Marchesiello, L. Šnobl. Superintegrability of separable systems with magnetic field: the cylindrical case. *J. Phys. A* **54** (2021) 425204

S. Bertrand, O. Kubů, L. Šnobl. On superintegrability of 3D axially-symmetric non-subgroup-type systems with magnetic fields. *J. Phys. A* **54** (2021) 015201

O. Kubů, L. Šnobl. Superintegrability and time-dependent integrals. *Archivum Math.* **55** (2019) 309–318

SELECTED TALKS

Full list: ondrejkubu.github.io.

Invited talk, *Geometry and Integrable Systems*, La Trobe University, Melbourne, Australia (Feb 2026)

Invited talk, *Superintegrability, Exact-Solvability, and Representation Theory* (Nov 2024)

Seminar, Università degli Studi di Milano, Italy (Feb 2026)

XXXIII International Fall Workshop on Geometry and Physics (Sep 2025)

XXXIV International Colloquium on Group Theoretical Methods in Physics (GROUP34) (Jul 2022)